EMIL VU

kietvu112yh@gmail.com | linkedin.com/in/emilvu112 | emilvu.github.io

EDUCATION

University of Victoria **Victoria, BC**
Master of Electrical and Computer Engineering (MEng*)* *Expected: Sep. 2026*
Bachelor of Electrical Engineering (BEng), with Co-ops *Sep. 2019 - Sep. 2025*

WORK EXPERIENCE

University of Victoria **Victoria, BC**
Graduate Teaching Assistant (ECE 250, ECE 299, ECE 350, ECE 355, ECE 441, ECE 455) *Sep. 2025 - Aug. 2026*

- Validated VHDL designs on Xilinx Vivado / Digilent Nexys A7 FPGA-adder/multiplier logic, a VGA image-processing pipeline, and a MicroBlaze soft-core + AXI system debugged via ILA.
- Operated SDR labs in GNU Radio on USRP/RTL-SDR (AM/SSB/FM/PSK, FLEX decoding) and analog-circuits labs on the oscilloscope, function generator, DMM, and wattmeter.
- Programmed and debugged FreeRTOS firmware on STM32F4 (tasks, queues, software timers) and a custom EDF deadline-driven scheduler.
- Programmed and debugged bare-metal STM32F0 in C/CMSIS-timers, EXTI interrupts, DAC/ADC, and an SPI OLED, with NE555/optocoupler PWM verified on a function generator, oscilloscope, and DMM.

Avigilon - Motorola Solutions **Vancouver, BC**
System Test & Validation (Co-op) *May. 2023 - Dec. 2023*
Product: Advanced security solutions with high-definition surveillance cameras and access control systems.

- Developed a new Python test automation framework, improving automation efficiency by 30% and reducing testing time by 8 hours per cycle.
- Designed and deployed 3D-printed testing jigs using Fusion 360, reducing cost by $300 per station.
- Created test specifications according to system requirements. Automated test development, maintenance, performance, and execution for software and hardware products.
- Integrated test automation with CI/CD pipeline using Jira, Bamboo, and Docker within Agile.

Loop Energy **Burnaby, BC**
Electrical Engineer (Co-op) *Sep. 2022 - Apr. 2023*
Product: High-efficiency hydrogen fuel cell systems and batteries for commercial vehicles.

- Assembled 10+ wiring harnesses and ran first-article inspections with root-cause analysis, driving corrective actions that improved design quality and accelerated bring-up.
- Built a Raspberry Pi 4 + PiCAN3 Vehicle Control Unit (VCU) simulator in Python + CAN to validate controller behavior, remove physical test benches and shorten validation by days.
- Co-designed a Fuel Cell Unit (FCU) simulator with swappable wiring and I/O access to test harnesses, sensors, and pumps; enforced controller-spec safety constraints and streamlined testing.

PROJECTS

FOC Firmware for Adaptive Arm-Rehabilitation Device **University of Victoria**
M.Eng. Thesis Project *Sep. 2025 - Present*

- Built a complete Field-Oriented Control (FOC) firmware from scratch in C/C++ on an 8-bit ATmega328P (Arduino UNO): Clarke/Park transforms, min–max SVPWM, register-level 31.4 kHz PWM, ISR-based quadrature decoding, and anti-windup PID.
- Added automated identification of motor parameters (R, L, flux linkage) and a closed-loop impedance controller, then validated a programmable spring–damper rehabilitation mode with a DRV8302 inverter, AMT103 encoder, GM4108 gimbal motor, and a 3D-printed Fusion 360 mount/pendulum.

Portable HF Linear Amplifier with BPF/LPF for Emergency Communications **University of Victoria**
B.Eng. Capstone–Group Project *Sep. 2024 - Apr. 2025*

- Co-designed a 140 W PEP linear HF amplifier (1.8-30 MHz, AM/SSB) based on the Motorola EB63, modernizing obsolete components and adding mode select, manual bypass, and fuse circuits.
- Designed and simulated a two-stage π-network constant-k bandpass filter (1.8-30 MHz) and a 9th-order T-network Chebyshev low-pass filter (30 MHz) in LTspice; produced schematics and PCB layouts in KiCAD.
- Assembled the prototype with a custom coil-winding jig and a laser-cut/3D-printed enclosure; validated amplifier gain (AM-mode power sweep) and filter S21 response against spec on a VNA.

SKILLS

Languages: Python, C/C++, Verilog, VHDL, MATLAB.
Embedded & Hardware: FreeRTOS, STM32, BLDC/PMSM motor control, FOC, PWM, CAN, SPI, I2C, UART.
Software & DevOps: Git, Jira, Docker, CI/CD, Bamboo, Agile
RF & Analog: HF amplifier design, filter design (Chebyshev, pi-network), LTspice.
CAD/Design: SolidWorks, Fusion 360, KiCAD, Eagle PCB.